(ii) Let $y = (\log x)^{\cos x}$

Taking logarithm on both sides

$\log y = \cos x \ \log(\log x)$

Differentiating with respect to $x$,

$$\frac{1}{y}\cdot\frac{dy}{dx} = \cos x\frac{1}{\log x}\cdot\frac{1}{x} + \left[\log(\log x)\right](-\sin x)$$

$$\frac{dy}{dx} = y\left[\frac{\cos x}{x\log x} - \sin x \ \log(\log x)\right]$$

$$= (\log x)^{\cos x}\left[\frac{\cos x}{x\log x} - \sin x \log(\log x)\right]$$

**Example 5.41**

If $x^y = e^{x-y}$, then prove that $\frac{dy}{dx} = \frac{\log x}{(1+\log x)^2}$

***Solution***

$x^y = e^{x-y}$

Taking logarithm on both sides,

$y \log x = (x-y)$

$y(1+\log x) = x$

$y = \frac{x}{1+\log x}$

Differentiating with respect to $x$

$$\frac{dy}{dx} = \frac{(1+\log x)(1) - x\left(0+\frac{1}{x}\right)}{(1+\log x)^2}$$

$$= \frac{\log x}{(1+\log x)^2}$$

**Example 5.42**

Differentiate: $\sqrt{\frac{(x-3)(x^2+4)}{3x^2+4x+5}}$

***Solution***

Let $y = \sqrt{\frac{(x-3)(x^2+4)}{3x^2+4x+5}}$

$$= \left[\frac{(x-3)(x^2+4)}{3x^2+4x+5}\right]^{\frac{1}{2}}$$

Taking logarithm on both sides,

$$\log y = \frac{1}{2}\left[\log(x-3) + \log(x^2+4) - \log(3x^2+4x+5)\right]$$

$\left[\because \log ab = \log a + \log b \text{ and } \log\frac{a}{b} = \log a - \log b\right]$

Differentiating with respect to $x$,

$$\frac{1}{y}\cdot\frac{dy}{dx} = \frac{1}{2}\left[\frac{1}{x-3} + \frac{2x}{x^2+4} - \frac{6x+4}{3x^2+4x+5}\right]$$

$$\frac{dy}{dx} = \frac{1}{2}\sqrt{\frac{(x-3)(x^2+4)}{3x^2+4x+5}}\left[\frac{1}{x-3} + \frac{2x}{x^2+4} - \frac{6x+4}{3x^2+4x+5}\right]$$

## Exercise 5.7

1. Differentiate the following with respect to $x$,

   (i) $x^{\sin x}$ (ii) $(\sin x)^x$

   (iii) $(\sin x)^{\tan x}$

   (iv) $\sqrt{\frac{(x-1)(x-2)}{(x-3)(x^2+x+1)}}$

2. If $x^m \cdot y^n = (x+y)^{m+n}$, then show that $\frac{dy}{dx} = \frac{y}{x}$

### 5.3.5 Differentiation of parametric functions

If the variables $x$ and $y$ are functions of another variable namely $t$, then the functions are called a parametric functions. The variable $t$ is called the parameter of the function.

If $x = f(t)$ and $y = g(t)$, then

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$



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